

Description	Owners/Teams	Impact	Effort (FTE)	Milestones (M) and Decision Points (D) (year: description)	Risks and External Dependencies	Prob. of Applicability /Success/Adoption (Low, Medium, High)
Data Processing Framework and Programming Model						
2.1 Development and maintenance of the CMSSW event processing Framework	Development team, CMS ownership: Core Software (Matti Kortelainen, Shahzad Muzaffar)	Throughput improvement: optimal use of many-cores CPUs, heterogeneous computing	3.5	'19-'34 M: keep up-to-date with evolving libraries, support MT platforms (100's of cores), Identify more places for concurrency, improve I/O for more concurrency, improve support for heterogeneous computing	Dependencies: core software libraries (TBB, ROOT) Risks: I/O bottleneck for scaling, divergences in the core software libraries.	Success: High
2.7 Services for optimized network ML inference on coprocessors (SONIC)	Development team: CMS CMS ownership: Core Software (Matti Kortelainen, Shahzad Muzaffar + Kevin Pedro), ML (J.R. Vlimant, G. Kasieczka, J. Ngadiuba)	Speed up ML inference by factors of 10–100 or more with respect to local CPU.	2.5	'23 (M): conclude local developments (adding features, porting algorithms) '23 (M): perform tests within CMS production '24 (M): conclude integration w/ CompOps, PdmV, etc. '24 (M): large-scale deployment of workflows in official infrastructure '25 (D): decision on adoption for HL-LHC	Dependencies: core framework (ExternalWork) Nvidia Triton server Container software (Docker, Singularity) Other stakeholders: DPGs, POGs (algorithms), CompOps, PdmV (production) Risks: coprocessor access Resource usage (network, memory)	Medium
2.4 Batching event data in the framework	Development team, CMS ownership: Core Software (Matti Kortelainen, Shahzad Muzaffar)	> 2x throughput on GPUs without concurrent kernel support	1.0 1.0	'22-'23 (D): demonstrate performance improvement with respect to concurrent CUDA streams '24 (M) Prototypes of different candidate batching mechanisms	Dependencies: framework, portability tools Risks: performance improvements not worth the added complexity.	Adoption: Core Software L2 say Medium (L1s say High)
2.2 Performance portability	Development team: community-wide (CERN, US-CCE) with strong CMS component, including HLT and offline CMS ownership: Core Software (Matti Kortelainen, Shahzad Muzaffar)	Enable GPU algorithm development by improving productivity	2.5 2.5	'22 D: Alpaka was selected as portability tool for Run 3 '22-'24 M: investigate alternative portability tools as potential replacement for Alpaka in HL-LHC '25 D: decision on portability tool for HL-LHC	Dependencies: portability libraries, HW vendor native technologies. Risks: HW vendors not providing a common solution, native technologies becoming incompatible with each other.	Adoption of a tool other than Alpaka in HL-LHC: Medium
Computing Infrastructure						
3.16 Submission Infrastructure: Support for Heterogeneous Resources including non-x86.64 CPUs	Development team: CMS CMS ownership: DRP (D. Spiga, C. Wissing)	Access HPC resources at HPCs and modernized WLCG centers (~10% more CPUs)	0.5 0.5 0.5	'22 M: finalize POWER9 as 1st non-x86 architecture for production '23 M: GPU workflows available for production in the distributed infrastructure '23 M: Support for other non-x86 architecture, e.g. ARM	Dependencies: HTCondor Risks: insufficient expertise in SI and DMWM.	Success: High
3.13 Benchmarking of heterogeneous platforms	Development team: WLCG/HPC Integration CMS ownership: Upgrade R&D Maria Girone	Estimate site value by use of computing performance benchmarking suites for HPCs	0.1 0.1	'21 M: HPC benchmarking capabilities added '22-'23 M: reco/sim, including AI, workflows done. CMS component of HEPScore benchmark done.	Dependencies: benchmarking framework is done in collaboration with WLCG, HEPiX, EGI-ACE	Success: High